Session Objectives:

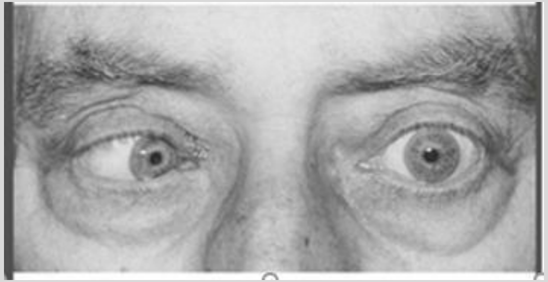
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

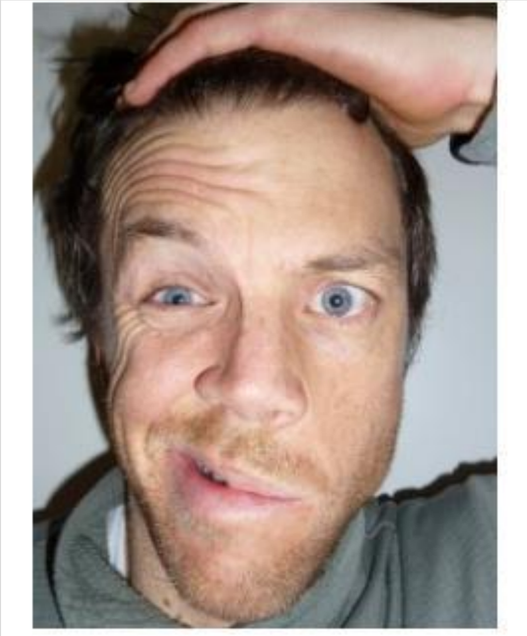
- Learn how to obtain a health history in a patient with a neurological complaint
- Assess Mental Status
- Review the Cranial Nerves
- Learn how to perform a Cranial Nerve examination
- Review the eye examination
- Correlate a neurological diagnosis from a Cranial Nerve examination
- Document physical findings for a Cranial Nerve examination and Eye examination

Why Perform a Neurological Exam?

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine





Components of a Screening Neurological Examination

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- **Mental Status**—level of alertness, appropriateness of responses, orientation to date and place
- **Cranial Nerves**
- Visual Acuity
- Pupillary light reflex
- Eye movements
- Hearing
- Facial strength- smile, eye closure
- Speech
- **Motor System**
- Strength—shoulder abduction, elbow flexion/extension, wrist extension, finger abduction, hip flexion, knee flexion/extension, ankle dorsiflexion
- **Gait**—casual and tandem
- **Coordination**—fine finger movements, finger-to-nose or finger to chin
- **Reflexes**- Deep Tendon Reflexes and Plantar responses
- **Sensory System**—one modality at toes—can be light touch, pain/temperature, or proprioception

Bates, 13th Edition, Chapter 24- American Academy of Neurology Guidelines for a Screening Neurologic Examination

Neurological Examination

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Start with taking the patient's history
- Do they have neurologic disease?
- If so, what is the localization of the lesion or lesions?
- Are the findings symmetrical?
- What is the pathophysiology of the abnormal findings?
- What is the preliminary differential diagnosis?

Glasgow Coma Scale

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Activity		Score
Eye Opening		
None	1 = Even to supraorbital pressure	
To pain	2 = Pain from sternum/limb/supraorbital pressure	
To speech	3 = Nonspecific response, not necessarily to command	
Spontaneous	4 = Eyes open, not necessarily aware	——
Motor Response		
None	1 = To any pain; limbs remain flaccid	
Extension	2 = Shoulder adducted and shoulder and forearm internally rotated	
Flexor response	3 = Withdrawal response or assumption of hemiplegic posture	
Withdrawal	4 = Arm withdraws to pain, shoulder abducts	
Localizes pain	5 = Arm attempts to remove supraorbital/chest pressure	
Obeys commands	6 = Follows simple commands	——
Verbal Response		
None	1 = No verbalization of any type	
Incomprehensible	2 = Moans/groans, no speech	
Inappropriate	3 = Intelligible, no sustained sentences	
Confused	4 = Converses but confused, disoriented	
Oriented	5 = Converses and is oriented	——
		TOTAL (3–15)*

A patient with a score of 3-8 is usually considered to be in a coma

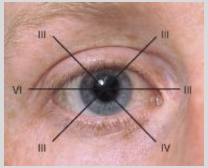
Bates, !3th Edition, Chapter 24- Table 24-13

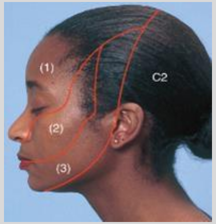
Summary of Cranial Nerves I-XII

Cranial Nerve	Function of Cranial Nerve	Abnormal Clinical Finding
CN I- Olfactory (Sensory)	Smell	Anosmia
CN II- Optic (Sensory)	Visual acuity, visual field, ocular fundus	Amaurosis
CN III- Oculomotor (Motor)	Ocular movement, lid elevation, pupil constriction, accommodation	Diplopia, ptosis, loss of accommodation
CN IV- Trochlear (Motor)	Extra-ocular movement, downward, internal rotation of eye	Diplopia
CN V- Trigeminal (Motor and Sensory)	Facial, scalp and teeth sensation, corneal reflex (sensory), jaw movement, voice/speech	Weakness of jaw, numbness of face
CN VI- Abducens (Motor)	Extra ocular movement, lateral deviation of the eye	Diplopia
CN VII- Facial (Motor and Sensory)	Taste, general sensation of palate larynx and external ear, lacrimal, submandibular and sublingual gland secretion, facial movements- including facial expression, closing the eye and closing the mouth	Dry mouth, paralysis of facial muscles, loss of lacrimation, loss of taste to anterior 2/3rd of tongue

Bates 12th Edition, Chapter 17

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine





Summary of Cranial Nerves I-XII

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Cranial Nerves	Function of Cranial Nerve	Abnormal clinical finding
CN VIII- Acoustic or Vestibulocochlear (Sensory)	Hearing, balance/equilibrium	Vertigo, tinnitus, deafness, nystagmus
CN IX- Glossopharyngeal (Motor and Sensory)	Taste, sensation of pharynx, and ear, palate elevation, gag reflex (sensory)	Loss of taste in posterior $1/3^{rd}$ of tongue, partial dry mouth, pharynx anesthesia
CN X- Vagus (Motor and Sensory)	Taste, sensory to pharynx, larynx, ear, swallowing, phonation; motor to palate, pharynx and larynx; gag reflex (motor); parasympathetic innervation of heart and abdominal viscera	Dysphagia, hoarseness, paralysis of palate
CN XI- Spinal Accessory (Motor)	Head, neck and shoulder movement, phonation	Weakness of head, neck, shoulder, hoarseness
CN XII- Hypoglossal (Motor)	Tongue symmetry and position, voice/speech	Weakness, tongue wasting

Bates 12th Edition Chapter 17

Swartz 8th Edition Chapter 21

CN I- Olfactory (Sensory)

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Ask the patient to close their eyes and occlude one nostril
- Test substance is brought close to the unobstructed nostril
- Patient is instructed to sniff and identify the substance
- Repeat on the other side

*Test substance must be non-irritating (ex: fresh ground coffee, lavender, cloves, vanilla beans)

From Cranial Nerves 3rd Ed. ©2010 Wilson-Pauwels, Stewart, Akesson, Spacey, PMPH-USA

Bates 13th Edition Chapter 24

The Examination of the Eye

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Visual Acuity- CN II (Optic)
- Visual Field by Confrontation- CN II
- Pupil Reaction/Response- CN III (Oculomotor)
- Extra-Ocular Muscle Movement- CN III (Oculomotor), IV (Trochlear), VI (Abducens)
- Anterior and Posterior Segments- Funduscopic Examination

Distance Visual Acuity Testing

- Nomenclature

Distance between the patient and the eye chart

Distance at which the letter can be read by a person with normal visual acuity

Normal: 20/20

Below Normal: 20/40 or higher

Common Refractive Errors

NEW YORK INSTITUTE OF TECHNOLOGY

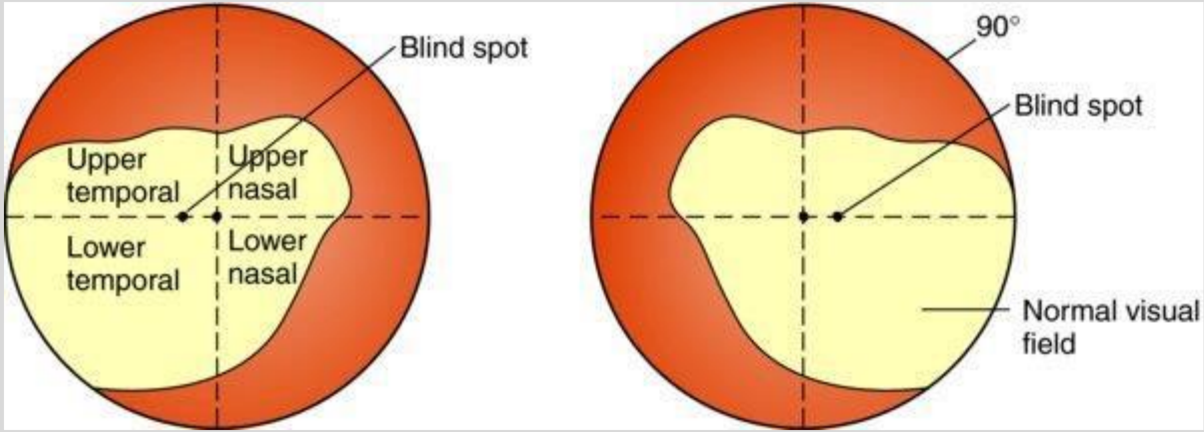
College of Osteopathic Medicine

Refractive error	Description
Hyperopia	Globe is too short. Converging lens is used to increase refractive power. Light rays focus behind the retina.
Myopia	Globe is too long. Light rays focus in front of the retina. Distant objects require diverging lens.
Astigmatism	Corneal surface is not perfectly spherical. Refractive power is different at different axes. Needs cylindrical corrective lens.
Presbyopia	Lens becomes unable to increase refractive power to accommodate on near objects.

Source: Ophthalmologic Evaluation. Vaughan & Asbury's General Ophthalmology, 18th edition, Chapter 2

Visual Fields

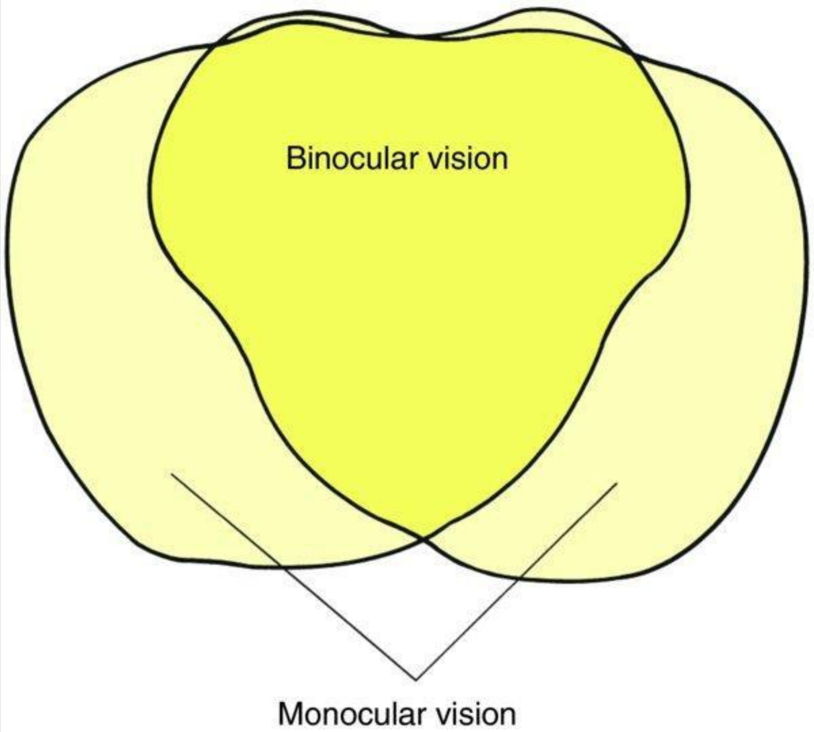
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



- Entire area seen by an eye when it looks at a central point
- The field extends farthest on the temporal side

Bates 13th Edition Chapter 12

Convergence of Visual Fields



Bates 12th Edition Chapter 7

Visual Field Defects

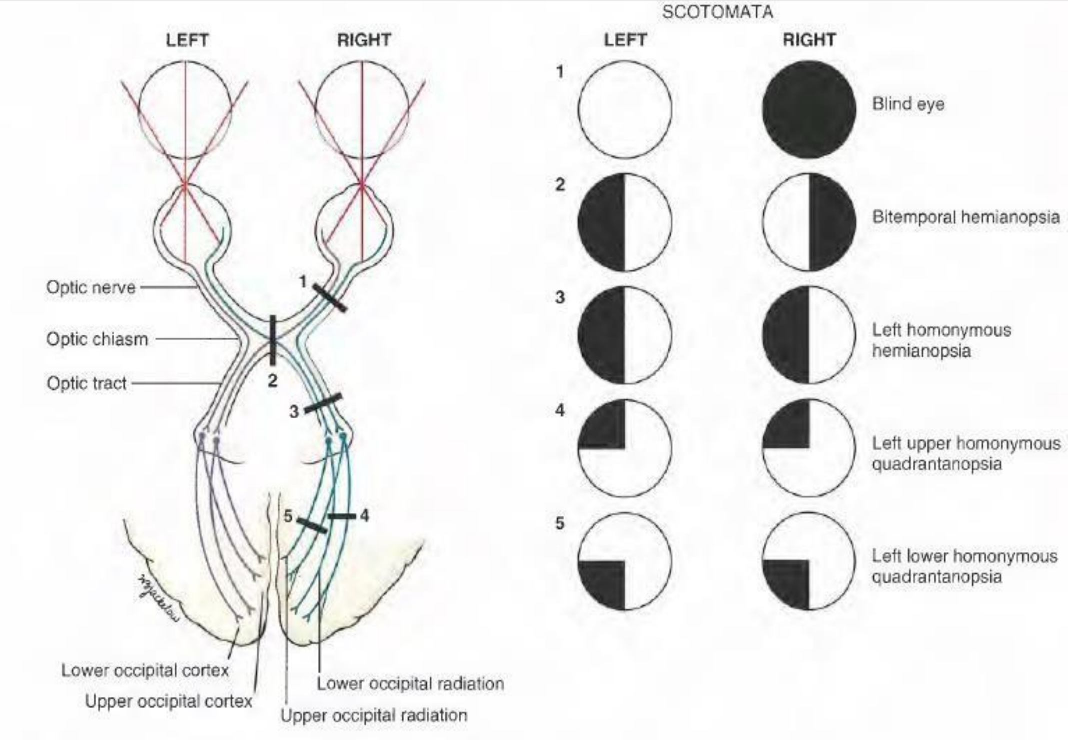


Figure 7–11. Visual field defects.

Swartz 7th Edition Chapter7

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Visual Fields by Confrontation CN II

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Static Finger Wiggle Test

- Have the patient look into your eyes.
- Place hands 2 feet apart lateral to patient's ears
- Wiggle both fingers and simultaneously bring them slowly forward as if a bowl. (curving inwards toward central vision)
- Have the patient inform you as soon as movement is seen.
- Repeat for all 4 quadrants of vision.

Bates 13th Edition Chapter 12

Visual Fields by Confrontation

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Further testing:

- If a defect is found with static finger wiggle test
- Test one eye at a time by having patient covering their eye with their hand.
- The examiner will cover their opposite eye as well.
- Have patient keep gaze with you and a point equal distant between the examiner and patient move wiggling fingers in the defect area until seen.
- Repeat steps in different levels to

  "map out" the defect

Bates, 11th Edition Chapter 7

Inspection

- Stand in front of the patient at eye level
- Assess symmetry, position of the eye and alignment
- Inspect the eyebrow- look for symmetry, sparing of hair
- Inspect the eyelids- ptosis, swelling, lesions, eversion or inversion of the lids

Bates, 13th Edition Chapter 12

Inspection

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Conjunctivitis

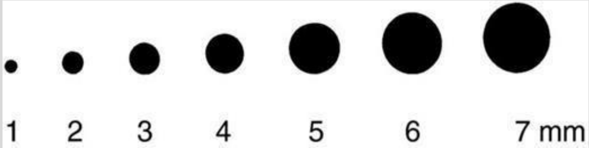
Exophthalmos

Ptosis

Assessment of Pupil

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Assess pupil size and shape
- Pupil sizes may be measured using a card with black circles of varying measurements from 1mm to 7mm.
- If >5mm - considered large
- If <3mm – small



Bates 13 Edition Chapter 12

Pupil Sizes

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Normal

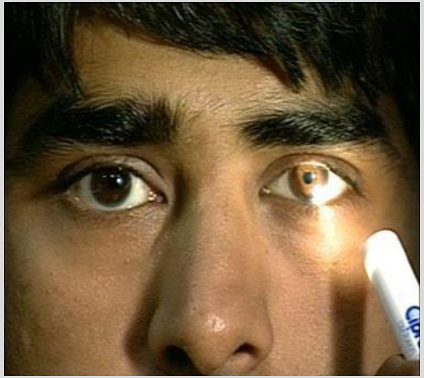
Dilated

Pupillary Response- CN III- Oculomotor (Motor)

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Direct penlight into eye while patient looking at a distance

- Confirm that the pupils respond to light
- Compare pupillary diameters



- Direct Response
  - Constriction of ipsilateral eye
- Consensual Response
  - Constriction of contralateral eye

Bates 13th Edition Chapter 12

Pupillary Response

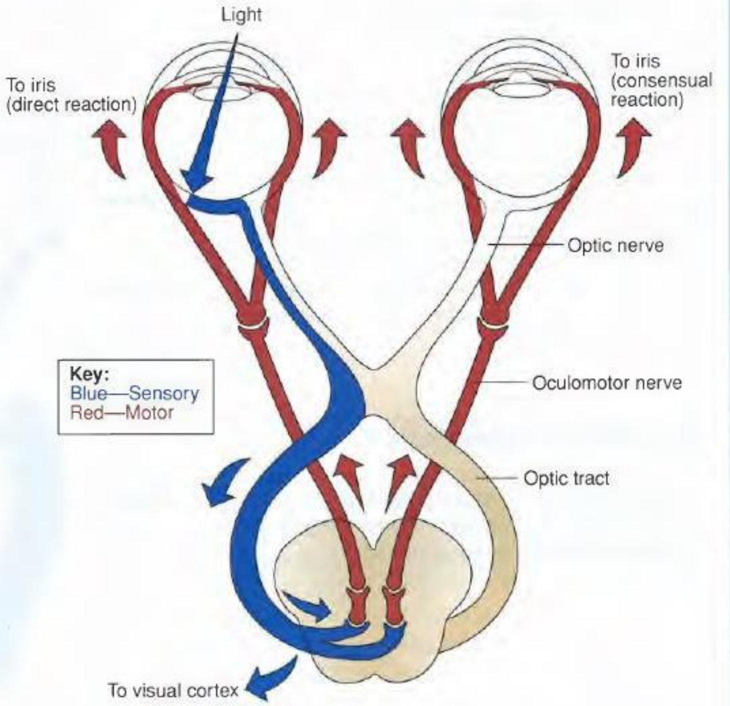


FIGURE 7-11. Pathways of the light reaction.

Bates 12th Edition Chapter 7

Near Reaction/Convergence

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Near reaction: Pupillary constriction with near effort (use if reaction to light is impaired or in question)

- In normal room light, hold a pencil or your finger 10 cm from the patient's eye.
- Ask the patient to alternate looking at an object on a far wall and then at the object in your hand.
- Watch for pupillary constriction with near effort.

Convergence: The ability of the two eyes to work synchronously together

- Ask the patient to follow your finger or object that is approximately 12 inches from the nose in toward 5-8 cm of the nose.
- Normal would be if the converging eyes can follow the object to that point.

Bates 13th Edition Chapter 12

Extraocular Movements CN III, IV, VI

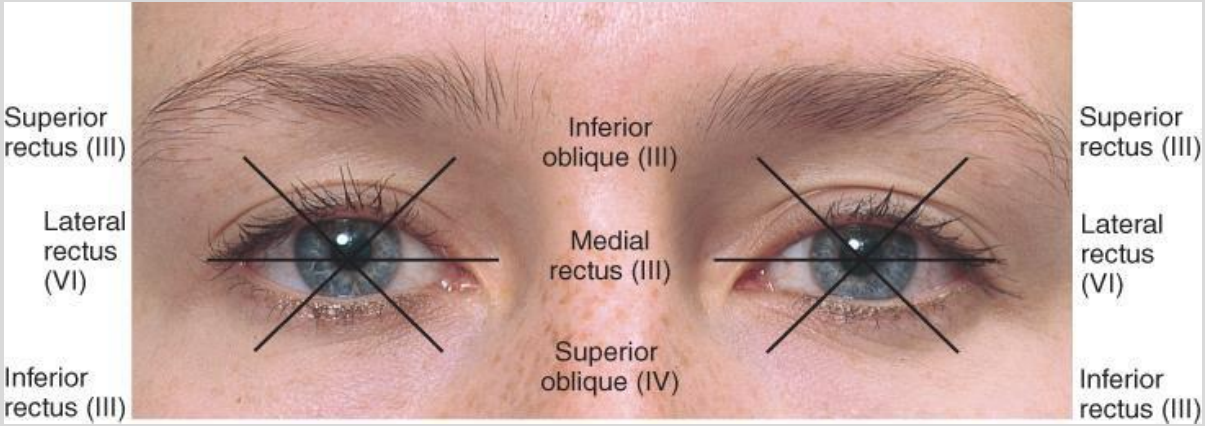
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

CN III- Oculomotor (Motor)

CN IV- Trochlea(Motor)

CN VI- Abducens (Motor)



Bates 13th Edition Chapter 12

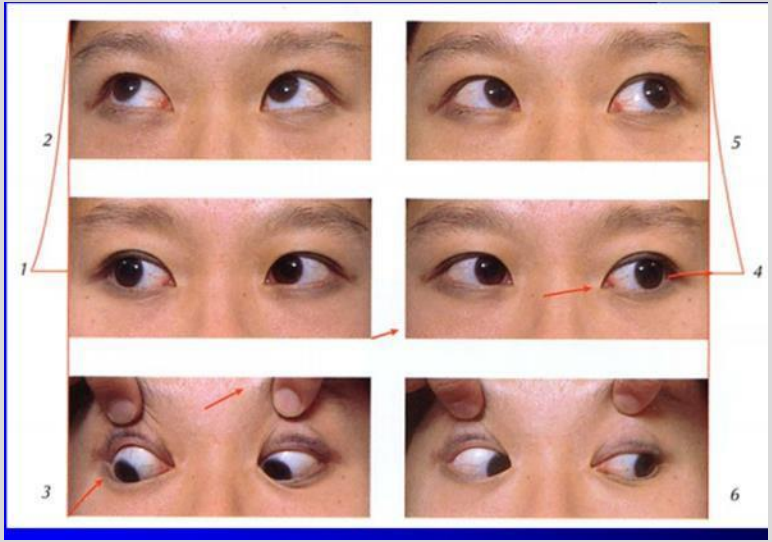
Extraocular Movement Testing

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

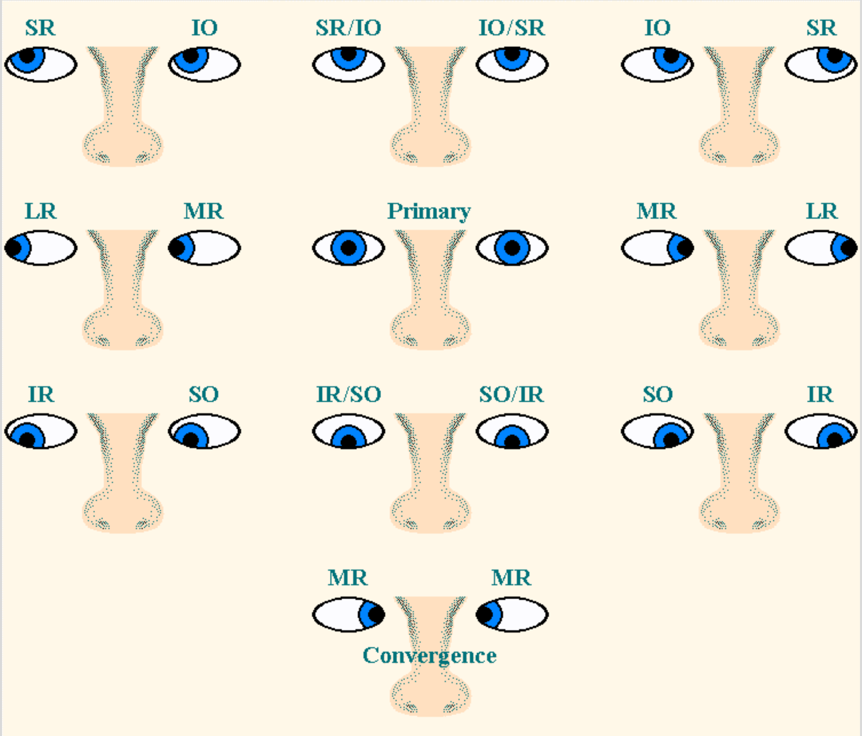
- Stand in front of the patient
- Ask patient to follow your finger with their eyes only, without turning or moving their head
- Make a wide "H" pattern in the air which covers all 6 cardinal directions of gaze

Bates 13th Edition Chapter 12



Ocular Motility

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



Source: Ophthalmologic Evaluation. Vaughan & Asbury's General Ophthalmology, 18th Edition, Chapter 2

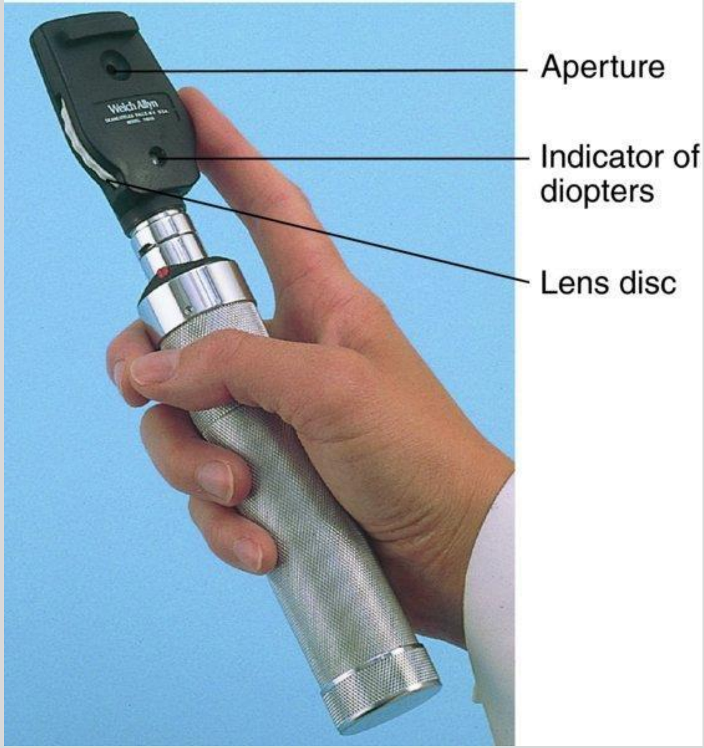
Diagnosis?

Swartz 8th Edition Chapter 10

Funduscopic Examination

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

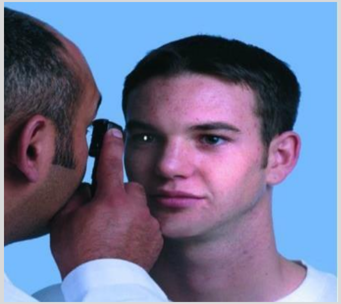
- Ophthalmoscope



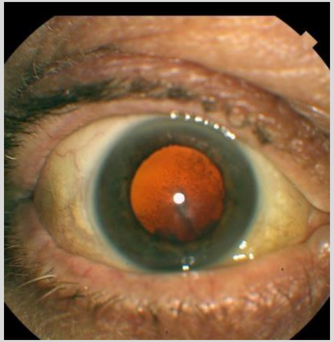
Fundoscopic Examination

New York Institute of Technology
College of Osteopathic Medicine

- Place your thumb of your free hand on the patient's eyebrow with the ophthalmoscope against your bony orbit



- When you shine the light into the patient's pupil you will see the orange glow of the pupil- Red Reflex



Fundoscopic Examination

- Continue to keep the light beam on the eye and continue the 15 degree angle until you are almost touching the patient's eyelashes.
- Locate the **Optic Disc** by following the blood vessels centrally and noting their enlargement as you approach the disc.
- Focus using the Lens Disc to bring the retina into focus

  If the patient is Myopic (nearsighted)- adjust the Lens Disc

  Counterclockwise (-) red numbers

  If the patient is Hyperopic (farsighted)- adjust the Lens Disc

  Clockwise (+) green numbers

Diopter Power Display→

Funduscopic Examination

- Inspect the **optic disc** for: clarity, color of disc and size of central physiologic cup
- Central physiologic cup is usually yellow-white

Bates 12 Edition Chapter 7

CN V-Trigeminal- Sensory

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Use a pin or cotton swab
- Assess at the three distributions of the Trigeminal Nerve: Ophthalmic,
  Maxillary
  Mandibular
- Test bilaterally

Corneal Reflex

- Out of the patient's line of vision, the cornea is touched lightly with a fine wisp of cotton
- Inspect for closure of bilateral eyelids (blinking) which is a normal response

Sensory limb of this reflex is with CN V

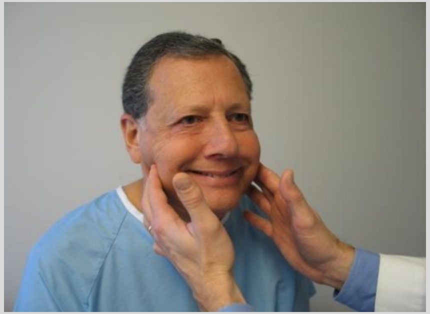
Motor response for both sides is with CN VII

Bates 13th Edition Chapter 24

CN V- Trigeminal -Motor

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Palpate the temporal and masseter muscles bilaterally and ask the patient to clench their teeth and note the strength of the contraction
- Have the patient open and move the jaw side to side



Bates 13th Edition Chapter 24

CN VII- Facial (Sensory and Motor)

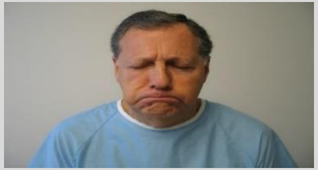
- Sensory- Taste from the Anterior two-thirds of the tongue
- Motor- Inspect the face for asymmetry

Ask patient :

Raise Eyebrows

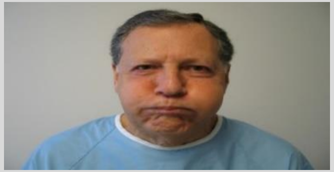
Close both eyes tightly- do not let me open them

Smile



Frown

Show teeth



Puff cheeks

Bates 13th Edition Chapter 24

Diagnosis?

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

CN VIII- Acoustic or Vestibulocochlear (Sensory)

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Gross Auditory Acuity

- Instruct patient to cover one ear
- Examiner whispers word or numbers into the uncovered ear from approximately 2 feet away
- Instruct patient to repeat the word/numbers
- Compare bilaterally

Alternative

- Instruct patient to cover one ear
- Examiner rubs thumb and forefinger together next to the uncovered ear and asks the patient if a sound was heard
- Compare bilaterally

Bates 13th Edition Chapter 13
Swartz 8th Edition Chapter11

Conductive vs Neurosensory Hearing Loss

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Weber Test

Test for lateralization of sound

- Use a tuning fork of 512 Hz
- Place vibrating tuning fork on top the patient's head
- Ask the patient to identify from which ear that they hear the sound.

Conductive versus Neurosensory Hearing Loss

Rinne Test- compare air conduction (AC) with bone conduction (BC)

- Use a 512 Hz tuning fork
- Strike the tuning fork to create a vibration
- The base of the vibrating tuning fork is placed against the mastoid process behind the ear that is being evaluated
- Ask the patient to inform you when they no longer hear the sound
- Move the tuning fork lateral to the patient’s ear without re-striking it
- Ask the patient if they hear the tuning fork and to inform you when they no longer hear it.

Bates 13th Edition Chapter 13

Rinne Test

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Normal- Air Conduction is greater than Bone Conduction
- Unilateral Sensorineural Loss- AC > BC
- Conductive Loss- BC= AC or BC> AC

CN IX and X

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

CN IX-Glossopharyngeal (Sensory & Motor)

Sensory- Gag reflex; Taste for posterior 1/3 of the tongue

CN X- Vagus (Sensory and Motor)

- Listen to the quality of the patient's voice
- Have the patient swallow and observe
- Instruct the patient open their mouth and say "ah"
  Observe for: symmetrical movement of the soft palate and location and movement of the uvula
- Check for the gag reflex by lightly stimulating the back of the throat using a tongue depressor on each side of the posterior pharynx.

Bates 13th Edition Chapter 24
Swartz 8th Edition Chapter 21

CN XI- Spinal Accessory (Motor)

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Observe and palpate for atrophy or fasciculations in the trapezius muscles bilaterally
- Ask the patient to shrug their shoulder against your resistance.

AND

- Place your hand on one side of the patient's face and ask the patient to turn their head towards that side.
- Apply resistance

Bates 13th Edition Chapter 24

CN XII- Hypoglossal (Motor)

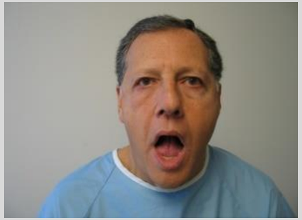
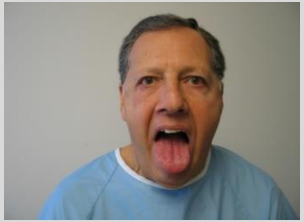
NEW YORK INSTITUTE OF TECHNOLOGY

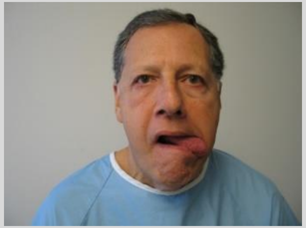
College of Osteopathic Medicine

- Instruct the patient to open their mouth
- Inspect the tongue as it lies on the floor of the mouth
- Instruct patient to protrude the tongue

  Note any asymmetry, atrophy or deviations

  from the midline
- Instruct the patient to move the tongue from side to side
- Have patient push the tongue against the inside of each cheek as the examiner palpates externally for strength




Bates 13th Edition Chapter 24
DeJong's 8th Edition Chapter 20

Diagnosis?

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine